Multimedia Tools and Applications (2023) 82:16985–17007
https://doi.org/10.1007/s11042-022-14164-5

Multi-objective Archimedes Optimization Algorithm with Fusion-based Deep Learning model for brain tumor diagnosis and classification

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Received: 7 February 2022 / Revised: 4 June 2022 / Accepted: 27 October 2022 /
Published online: 5 November 2022
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Abstract
Earlier identification of brain tumors (BT) is essential to increase the survival rate. Magnetic Resonance Imaging (MRI) is a commonly employed method that records brain abnormalities by the use of several modalities for clinical study. The recently developed computer vision and image processing schemes can be used for the detection and localization of tumor regions in the brain, which can be utilized for further treatment. In this regard, the study presents a novel Multiobjective Archimedes Optimization Algorithm with Fusion based Deep Learning (MOAOA-FDL) technique for brain tumor diagnosis and classification. In addition, the MOAOA-FDL technique preprocesses the MRI images via contrast enhancement and skull stripping. Moreover, AOA with Shannon entropy based multi-level thresholding approach is developed for medical image segmentation. Furthermore, the fusion of two deep learning models namely MobileNet and EfficientNet models is employed for feature extraction process. Finally, the AOA with long short term memory (LSTM) method is applied for classification model and thus allocates proper class label to it. The AOA is used to properly choose the hyper parameters like batch size, learning rate, and epoch count. The design of fusion process and MOAOA for BT diagnosis demonstrates the innovation of this study. For showcasing the better performance of the MOAOA-FDL method, a series of simulations have been executed utilizing benchmark dataset. The experimental outcome shows that the MOAOA-FDL method has outperformed the other recent approaches in terms of several performance measures.

Keywords Brain tumor · Image classification · Magnetic resonance imaging · Deep learning · Multiobjective optimization · Fusion models · Computer aided diagnosis

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1 Introduction

Brain tumor (BT) involves an abnormal cell aggregation that grows inside the brain tissue. The BT is classified into malignant tumor and benign tumor [3]. The benign tumor might be cured by surgical treatment, whereas malignant tumor is the deadliest type of cancer and might directly lead to death [14]. Also, BT is classified into metastatic BT metastasized from different body parts to the brain and primary tumor create from the brain or derive from the brain nerves [2]. The various symptom appears with distinct lesion regions like visual decline, headache, vomiting, confusion, and epilepsy. A complete classification splits BT into four grades, high the grade, a further malignant the BT. Based on the Global cancer analysis 2020 [24], there are around 308,000 novel cases of brain cancer in 2020, accounting for around 1.6% of each novel case of cancer, and around 251,000 death cases in brain cancer, accounting for around 2.5% of each cancer death. Earlier diagnosis is significant for the successful treatment of BT [13]. This imaging technique provided data on the size, shape, and position of BT, assisting doctors to develop treatment planning and make precise diagnoses. Magnetic resonance imaging (MRI) is the popular scanning method in neurology [5]. MRI employs radio frequency signals to stimulate the target tissues under the effect of stronger magnetic fields for producing an image of its internal structure. It has the advantage of zero ionizing radiation exposure and higher soft tissue contrast. Thus, MRI is better suited for the diagnosis of brain lesions [8].

For earlier BT classification and detection, a computer-aided diagnosis (CAD) could assist in the second opinion of the radiotherapist in the hospitals [22]. A CAD method comprises some major steps in a consecutive manner like feature extraction of pre-processed MRI scans, pre-processing of original MRI scans, classification using a supervised learning algorithm, and feature reduction. In the pre-processing stage, the original MRI scans are enhanced whereby tasks including noise removal and contrast enhancement are implemented. This is significant for classical or manual feature extraction. The traditional features are shape, color, texture, etc. However, these features are not effective for higher dimension datasets. Some authors proposed feature reduction techniques for the rapid implementation of CAD scheme [15]. But it is not an effective way because it drops important features and may neglect. The final feature is classified by supervised learning approaches namely Naïve Bayes, k-nearest neighbour (KNN), and so on [12]. This kind of CAD technique does not support multi-class classification problems. Lately, the entrance of deep learning (DL) in the area of medical imaging exhibited greater accomplishment for the classification task, particularly the multi-class classification problem with better performance. In this context, the DL algorithm is effectively employed for large dimension datasets.

This paper introduces an efficient multi-objective Archimedes Optimization Algorithm with Fusion based Deep Learning (MOAOA-FDL) model for brain tumor diagnosis and classification. The MOAOA-FDL technique involves the design of AOA with Shannon entropy based multi-level thresholding technique for medical image segmentation. Besides, the fusion of two deep learning models namely MobileNet and EfficientNet models is employed for feature extraction process. At last, the AOA with long short-term memory (LSTM) model is applied for classification process. In order to showcase the enhanced performance of the MOAOA-FDL technique, a series of

simulations were carried out using benchmark dataset. In short, the paper's contribution is summarized as follows.

- An intelligent MOAOA-FDL technique comprising of pre-processing, Shannon entropy based multi-level thresholding, entropy based fusion process, AOA with LSTM model for BT detection and classification using MRI images is presented. To the best of our knowledge, the MOAOA-FDL model has never presented in the literature.
- A novel AOA with Shannon entropy based multi-level thresholding approach is introduced by incorporating the concepts of AOA based optimal threshold value selection process.
- Employ entropy based fusion process involving MobileNet and EfficientNet based feature extractors.
- Hyperparameter optimization of the LSTM model using AOA algorithm using cross-validation helps to boost the predictive outcome of the MOAOA-FDL model for unseen data.

The rest of the paper is planned in such a way. Section 2 illustrates the related works and Section 3 elaborates on the proposed algorithm. Following, Section 4 shows the performance validation and Section 5 concludes.

2 Literature review

Rajasree et al. [16] developed a semantic-based U-Net model that explores a 3∗3 kernel size. The small kernel has an impact against over-fitting in the deep structure and offers a small amount of weight in this network. Multi-scale multimodal convolutional neural network (MSMCNN) using DL semantic segmentation approach is utilized for multimodality of MRI. Díaz-Pernas et al. [7] suggested a fully automatic BT classification and detection method with a DCNN which consists of a multiscale model. The major difference between the proposal regarding earlier work is that input image is processed in 3 spatial scales and processing pathways. This method is stimulated by the intrinsic process of the Human Visual System.

Sadad et al. [18] proposed segmentation by using Unet framework with ResNet50 as mainstream on the Figshare dataset and accomplished a level of 0.9504 of the intersection over union (IoU). The data augmentation and pre-processing concepts have been presented for enhancing the classification rate. The multiclass classification of BTs is implemented by evolutionary algorithm and reinforcement learning (RL) via transfer learning (TL) method. Also, Other DL approaches like DenseNet201, ResNet50, InceptionV3, and MobileNet V2 are employed. Sharif et al. [19] proposed a new automatic DL approach for classifying multi-class brain tumors. To realize the presented approach, the Densenet201 pre-trained DL method is finetuned and trained with a deeper transmission of imbalanced DL. The feature of the trained method is extracted from the standard pooling layer that characterizes deep data of all kinds of tumors. Khan et al. [10] developed a DL method for classifying brain tumors with region of interest data analysis for assisting practitioners. The presented approach contains three major stages: brain tumor classification with k-means clustering, pre-processing, and categorizing tumors into corresponding classes (benign or malignant) with MRI data by using a fine-tuned VGG19 architecture.

The authors in [20] proposed an effective model for brain MRI segmentation, called the HARIS-Heuristic technique. Initially, the MRIs are pre-processed by the use of adaptive contourlet transform for noise removal and a structural augmentation technique for skull removal process. Resultant MRIs are passed into the HARIS, to segment the texture by the use of multiobjective functions. In [21], the authors introduced a new AW-HARIS technique to segment the CT images of human liver effectively. The pre-processed image is passed into Anisotropic Weighted–-Heuristic Algorithm to segment them by the use of texture rlated details. It leads to accurate results with tolerable computation delay when compared to that of its counterparts.

Several existing works have used Kapur's function, Shannon's function, and Tsallis's function for image segmentation. Among these three models, Shannon's function performs well among the other techniques. Besides, many metaheuristics suffer from various limitations such as being stuck in local optima for solving global optimization problems, premature convergence, and trade-off between searching capabilities i.e., global and local search. Therefore, in this work, the AOA with Shannon's entropy technique is employed for BT segmentation. Besides, various hyperparameters have a significant impact on the efficiency of the DL model. Particularly, the hyperparameters such as batch size, learning rate, momentum, etc. are essential to attain effectual outcome. Since the trial and error method for hyperparameter tuning is a tedious and erroneous process, metaheuristic algorithms can be applied. Therefore, in this work, we employ AOA algorithm for the parameter selection of the LSTM model.

3 The proposed model

In this study, a new MOAOA-FDL technique has been developed for the detection and classification of BT using MRI images. The presented MOAOA-FDL technique encompasses different processes such as preprocessing, Shannon's entropy based segmentation, fusion enabled feature extraction, LSTM based classification, and AOA based hyperparameter optimization. Primarily, the input images are preprocessed to improve the contrast level and remove skulls. Followed by, the AOA with Shannon's entropy is applied for the segmentation of the images to determine the affected tumor regions. Then, the features are extracted and fused together by entropy process. Finally, the AOA-LSTM model receives the feature vectors and performs BT classification process. Figure 1 illustrates the overall process of MOAOA-FDL technique.

3.1 Image preprocessing

Initially, the input MRI image is preprocessed in two different ways i.e., skull stripping and contrast enhancement. For increasing the contrast level of the image, the contrast level adaptive histogram equalization (CLAHE) technique is applied. Then, skull stripping is needed to eliminate the skull from the background region. It is generally carried out by the use of imaging filter that separates the skull and remaining sections. In MRI images, the bone or skull regions have higher thresholding values (> 200) over tumors and other brain parts [9]. Therefore, the imaging filter can be used for separating the brain areas depending on the selected threshold values. Afterward, solidity property is applied for stripping the skull from brain MRI. Finally,

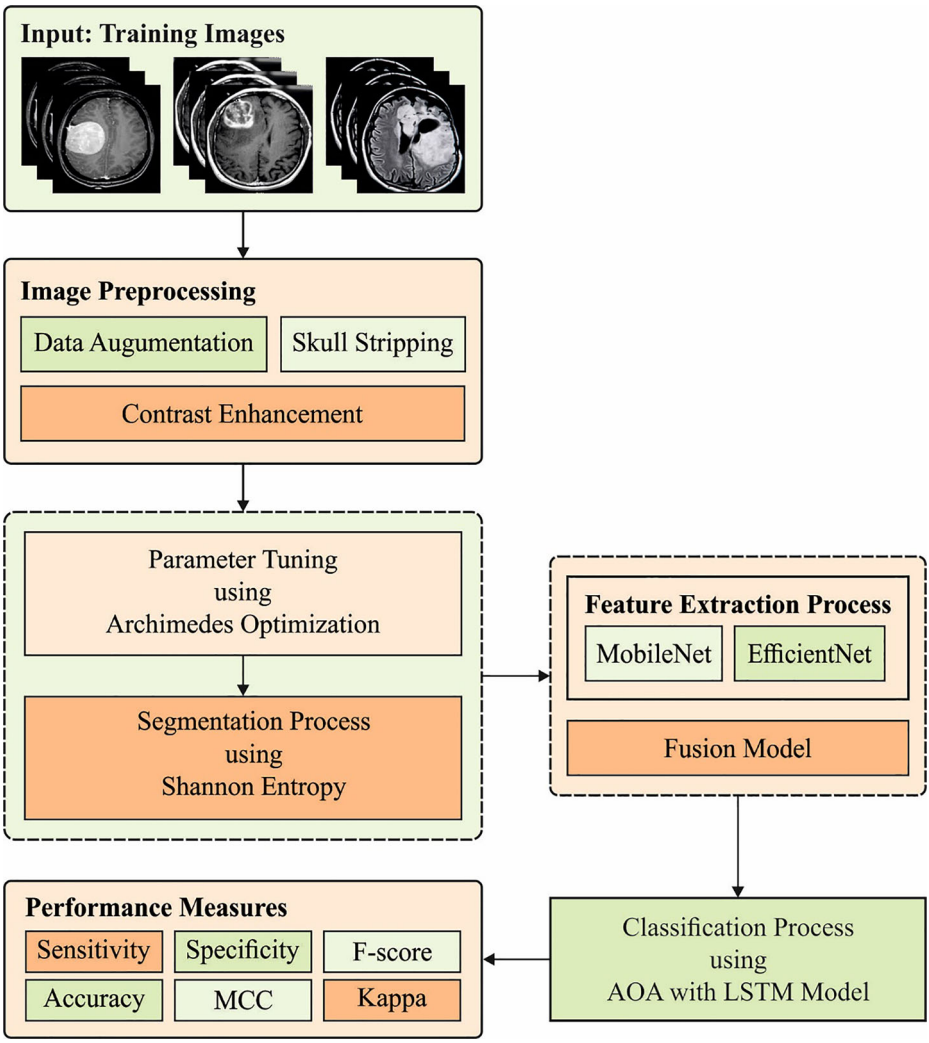

Fig. 1 Overall process of MOAOA-FDL technique

data augmentation method is carried out to increase the size of training data. In this work, data augmentation is implemented in four different ways namely flipping (horizontal and vertical) and rotation (90^0 and 180^0).

3.2 Overview of AOA

AOA is an optimization approach that is presented for tackling real time issues [1]. The motivation of optimizers is depending on the Archimedes principle state. It defines the behavior of the exerted force once an object is immersed partially or completely in the fluid. The optimization method initiates by presenting the first set of arbitrary particles (solution or object or position). All the particles have their acceleration, volume, and

density that are upgraded in a round methodology. The updated procedure of the AOA is shown below: compared with other optimizers, the AOA is categorized as global one whereby it considers the problem of exploitation and exploration. The AOA includes the following steps:

Step 1 Initialization: Proposed the first set of locations for the entire population:

$$O_i = lb_i + rand \times (ub_i - lb_i); i = 1, 2, \ldots, \mathrm{N} \tag{1}$$

Where as O_i indicates the ith solution in a population set of N solutions. lb_i and ub_i denotes the searching space lower and upper limits, correspondingly. The density (den) and volume (vol) initialization for ith solution is given below:

$$den_i = rand, vol_i = rand \tag{2}$$

In which $rand$ denotes an arbitrary vector of Dim dimension with value within [0, 1]. The first acceleration (acc) of the ith solution is given below [21]:

$$acc_i = lb_i + rand \times (ub_i - lb_i) \tag{3}$$

In this phase, the objective (fitness) function is estimated and the solution using optimal value is carefully chosen. The optimal value represents the maximal or the minimal for the maximization or minimization problem, correspondingly. Assign x_{best}, den_{best}, vol_{best}, and acc_{best}.

Step 2 Updating density, volume: The volume and density of a solution i in a round $t + 1$ are upgraded as follows:

$$\begin{aligned} den_i^{t+1} &= den_i^t + rand \times (den_{best} - den_i^t) \\ vol_i^{t+1} &= vol_i^t + rand \times (vol_{best} - vol_i^t) \end{aligned} \tag{4}$$

Whereas, den_{best} and vol_{best} denotes the volume and density of the global solutions, and $rand$ indicates an arbitrary value within 0 to 1.

Step 3 Transfer operator (TF) and density factor: The procedure initiates with collisions among objects for a while then they tend to attain equilibrium state. Thus, a TF is utilized for switching from the exploration stage to the exploitation stage:

$$TF = \exp\left(\frac{\mathrm{t} - \mathrm{t}_{\max}}{\mathrm{t}_{\max}}\right) \tag{5}$$

In which, the TF is increasing slowly with time till it is set at 1. t and $t_{\max}$ denotes round index and maximal number of rounds, correspondingly. As well, d assists in the global searching since its value is reduced with round t:

$$d^{t+1} = \exp\left(\frac{\mathrm{t}_{\max} - \mathrm{t}}{\mathrm{t}_{\max}}\right) - \left(\frac{\mathrm{t}}{\mathrm{t}_{\max}}\right) \tag{6}$$

Since d^{t+1} is slowly reducing, it assists the solution to converge in the potential searching region. The appropriate settings of this parameter get affected the tradeoff between the exploitation and exploration stages.

Step 4.1: Exploration stage: The object is taken into account in a collision state when $TF \leq 0.5$. Here, a material (mr) is randomly chosen and the object acceleration in the round $t + 1$ is upgraded by:

$$acc_i^{t+1} = \frac{den_{mr} + vol_{mr} \times acc_{mr}}{den_i^{t+1} \times vol_i^{t+1}} \tag{7}$$

In which n_i, vol_i, and acc_i denotes the i-th object acceleration, density, and volume, individually. While vol_{mr}, acc_{mr} and den_{mr} denotes the raworth material volume, acceleration, and density, correspondingly. It is noteworthy that $TF \leq 0.5$ ensures an exploration stage on at least one third of the searching method.

Step 4.2: Exploitation stage: Here, the object is taken into account as no collision state when $TF > 0.5$. The object acceleration is upgraded at the round $i + 1$ using Eq. (8):

$$acc_i^{t+1} = \frac{den_{best} + vol_{best} \times acc_{best}}{den_i^{t+1} \times vol_i^{t+1}} \tag{8}$$

where acc_{best} indicates the acceleration of optimal object.

Step 4.3: Normalized speed: For defining the % of variation, the normalization of acceleration takes place using Eq. (9):

$$acc_{i-norm}^{t+1} = u \times \frac{acc_i^{t+1} - min(acc)}{\max(\mathrm{acc}) - \mathrm{m}\,\mathrm{in}(\mathrm{acc})} + l \tag{9}$$

where, u and l indicate the maximum and minimum regularization ranges with the values of 0.9 and 0.1, correspondingly.

Step 5: Upgrade location: when $TF \leq 0.5$ (exploration stage), the ith object location is upgraded for the following iteration $i + 1$ as follows:

$$x_i^{t+1} = x_i^t + C_1 \times rand \times acc_{i-norm}^{t+1} \times d \times (x_{rand} - x_i^t) \tag{10}$$

Now, C_1 denotes a constant value of 2. Or else, when $TF > 0.5$ (exploitation stage), the solution is updated for the location as follows.

$$x_i^{t+1} = x_{best}^t + F \times C_2 \times rand \times acc_{i-norm}^{t+1} \times d \times (T \times x_{best} - x_i^t) \tag{11}$$

whereas C_2 denotes a constant value of 6.

Step 6 Calculation: The objective function f is estimated for the presented solution and the existing global optimal iteration is defined. Allocate acc_{best}, x_{best}, den_{best}, and vol_{best}.

3.3 Image segmentation

Next to image preprocessing, the segmentation of images takes place using the AOA with Shannon entropy function. Usually, entropy is adapted for computing the ambiguity of a parameter. In Paul and Bandyopadhyay's study, comprehensive data about Shannon entropy can be found [17]. The process is given in the following: consider selecting an image frame of size $M^{*}N$. Next, the gray value of pixel with coordinate (x, y) is formulated by (x, y), for $x \in \{1, 2, \mathrm{M}\}$ and $y \in \{1, 2, \mathrm{N}\}$. Where the set of each gray level $\{0, 1, 2, \ldots /\mathrm{L} - 1\}$ is represented as G and Lrepresents the number of gray levels of image I_0:

$$f(x, y) \in G \forall (x, y) \in image \tag{12}$$

The normalized histogram of image is signified as $H = \{h_0, h_1, \ldots, h_{L-1}\}$. For multi-thresholding issues, Eq. (12) is formulated by;

$$H(T) = h_0(t_1) + h_1(t_2), \ldots, h_{L-1}(t_{k-1}) \tag{13}$$

$$T* = \mathrm{Tmax}\{H(T)\} \tag{14}$$

Where $T*$ denotes the optimum threshold. Shannon's entropy must fulfill the above equation. In addition, the AOA is applied for the selection of optimum threshold values.

3.4 Fusion based feature extraction

During feature extraction process, the fusion of the features derived by the EfficientNet [23] and MobileNet [17] models takes place. The MobileNet and EfficientNet are used in this work due to the advantages of reduced network size, less number of parameters, and faster in performance. EfficientNet model has been developed for enhancing the results of the CNN with scaling in 3 dimensions namely width, depth, and resolution by the use of predefined scaling coefficients [23]. The EfficientNet model holds eight different versions namely B0 to B7. It makes use of transfer learning to reduce computation time and power. In this work, EfficientNet-B0 model is applied, which comprises 18 convolutional layers, D = 18, and every layer is related to a kernel $k(3, 3)ork(5, 5)$. The input images encompass R, G, and B color channels, everyone denotes a size of 224 × 224. The subsequent layers are scaled for reducing the size of the feature map. Next, the MobileNet network is presented for improving the real time outcomes of the DL model with restricted hardware constraints [4]. It can minimize the parameter count with no compromise in accuracy. Earlier works have reported that the MobileNet requires 1/33 of the variables of VGG-16 in accomplishing identical classifier outcomes. Conv_Dw_Pw is a separable and deep convolutional model, which comprises (Dw) and (Pw) depthwise and pointwise layers. The Dw are deep convolution layers with 3 × 3 kernels, whereas Pw is a commonly available convolution layer with 1 × 1

kernel. Every convolutional outcome can be considered via the batch normalization approach and rectified liner unit (ReLU) activation function.

Data fusion is applied in the applications of computer vision (CV) and machine learning (ML) techniques. Feature fusion is a major task that incorporates more than one feature vectors. The suggested technique depends on feature fusion and the feature vector is formulated as follows.

$$f_{EFF1\times n} = \{EFF_{1\times 1}, EFF_{1\times 2}, EFF_{1\times 3}, \cdots, EFF_{1\times n}\} \tag{15}$$

$$f_{\mathrm{MN1\times p}} = \{MN_{1\times 1}, MN_{1\times 2}, MN_{1\times 3}, \cdots, MN_{1\times p}\} \tag{16}$$

Furthermore, extracted feature is fused into an individual vector.

$$Fused(featuresvector)1 \times \mathrm{q} = \sum_{\mathrm{i}=1}^{2} \mathrm{f}_{\mathrm{EFF1\times n}} \times \mathrm{f}_{\mathrm{MN1\times p}} \tag{17}$$

Whereas f is fused vector (1×1186). The entropy is exploited to feature vector for choosing better features as per the score. The FS process is described mathematically as follows. Entropy is exploited for choosing 1186 score-based features from 7835 features.

$$B_{He} = -NHe_b \sum_{i=1}^{n} p(f_i) \tag{18}$$

$$F_{select} = B_{He}(\max(\mathrm{f_i}, 1186)) \tag{19}$$

Here, He indicates entropy and p denotes feature probability. Lastly, the chosen features are fed to the classifier for discriminating the glioma and healthy images.

3.5 Image classification

At the final stage, the derived feature vector is passed into the LSTM model BT detection and classification [11]. LSTM is a variant of recurrent neural network (RNN) that can be able to remember the information for extended periods. A major benefit of using LSTM networks lies in the fact that it resolves the vanishing gradient issue which makes network training tedious for a long sequence of words or integers. The gradients can be used to update the RNN parameters and for a long data; these gradients become smaller and smaller to the extent that, effectively, no network training can take place. LSTM networks help to resolve the issue and make it possible in capturing long-term dependencies among integers in sequences that are separated by a large distance.

The interior construction of LSTM encompasses forget output, and input gates, whereby all the gates process the input from the preceding gate and transmit them to the subsequent gate thereby controlling the data flow toward the last output. Usually, each gate can be controlled by *tanh* activation or sigmoid function, for instance, input gate i_t takes the responsibility of upgrading the dataset. The forget gate process input dataset from the i_t input gate and C_{t-1} the preceding cell state, it removes the dataset from the C_t existing state as required. The 0_t output gate forwards the last output to the subsequent LSTM units and holds output values for the sequential prediction. At the same time, C_t recurrent unit evaluates the C_{t-1} state of prior cell and x_t current input values with *tanh* function. The h_t value is evaluated

by using the scalar product of 0_t and *tanh* of C_t. At last, the concluding output is accomplished by passing h_t to the *softmax* classifier. The mathematical formulation of the aforementioned gates is given below:

$$f_t = \Phi(\widehat{W} \cdot [h_{t-1'} x_t] + B_f) \tag{20}$$

$$i_t = \Phi(\widehat{W}_i \cdot [h_{t-1'} x_t] + B_i) \tag{21}$$

$$C_t = tanh\left(\widehat{W}_C \cdot [h_{t-1}, x_t] + B_C\right) \tag{22}$$

$$C_t = f_t x C_{t-1} + i_t x C_t \tag{23}$$

$$0_t = \Phi\left(W_0 \left[h_{t-1'} x_t\right] + B_0\right) \tag{24}$$

$$h_t = o_t x tanh(\Phi(C) \tag{25}$$

$$Output = softmax(h_t) \tag{26}$$

Besides, the AOA is used to properly choose the hyperparameters like batch size, epoch count, and learning rate. The AOA approach resolves a fitness function (FF) for obtaining enhanced classification performance. It describes a positive integer for demonstrating the better performance of solution candidate. In this case, the minimized classifier error rate is viewed as the FF is offered in Eq. (27). The worst solution gains a maximal error rate and better solution is a minimum error rate.

$$\begin{aligned} fitness(x_i) &= Classifier\ Error\ Rate(x_i) \\ &= \frac{number\ of\ misclassified\ images}{Total\ number\ of\ images} * 100 \end{aligned} \tag{27}$$

4 Performance validation

4.1 Implementation details

The proposed model is simulated using Python 3.6.5 tool with some packages namely tensorflow-gpu = = 2.2.0, pandas, opencv-python, simpleitk, scikit-image, scikit-learn,

matplotlib, seaborn, Pillow, prettytable, scipy, pyqt5, tqdm, and numpy. The proposed model is experimented on PC i5-8600k, GeForce 1050Ti 4GB, 16GB RAM, 250GB SSD, and 1 TB HDD. The parameter settings are given as follows: learning rate: 0.00001, dropout: 0.5, batch size: 8, and activation: ReLU.

4.2 Dataset details

For experimental validation, the MOAOA-FDL technique is tested using benchmark BT data set from Kaggle repository (https://www.kaggle.com/navoneel/brain-mri-images-for-brain-tumor-detection) and FigShare dataset (https://figshare.com/). The first Kaggle dataset holds 98 images under Normal class and 155 images under Tumor or Abnormal class. The image size ranges from 192*192 pixels to 630*630 pixels. Figure 2 demonstrates the sample images.

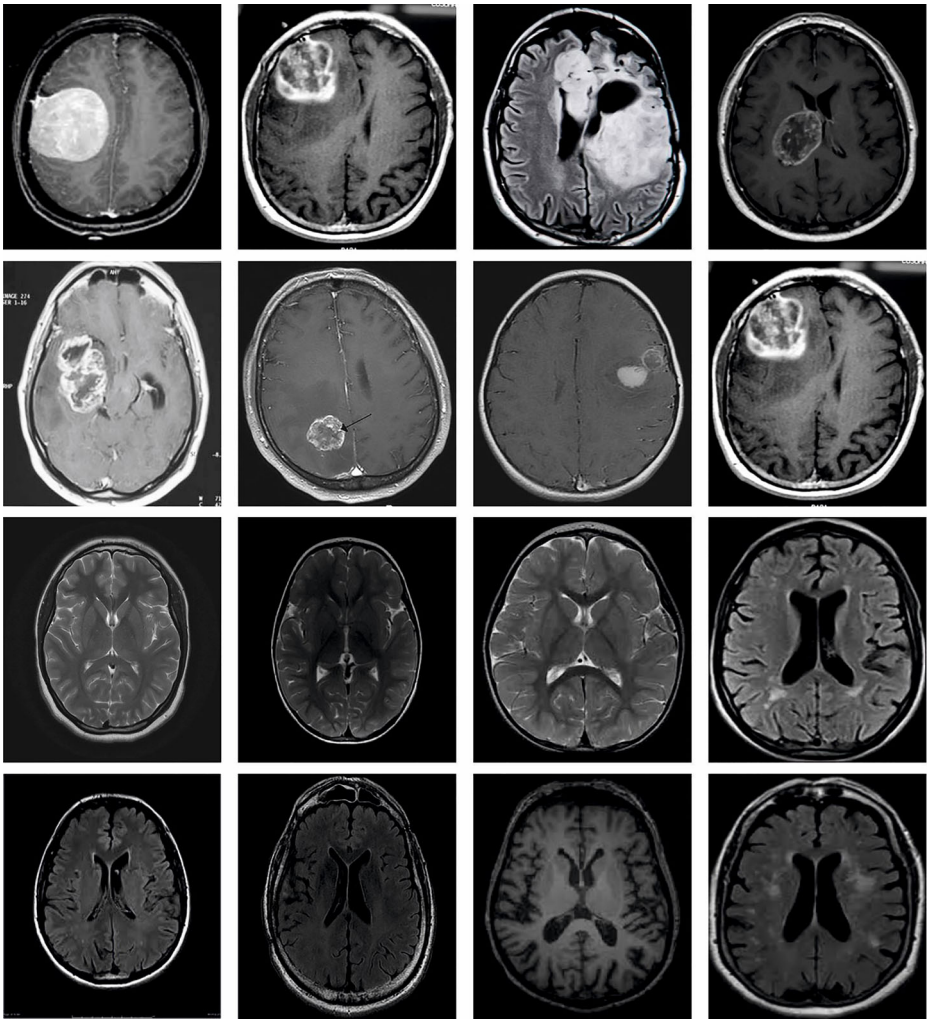

Fig. 2 Sample images

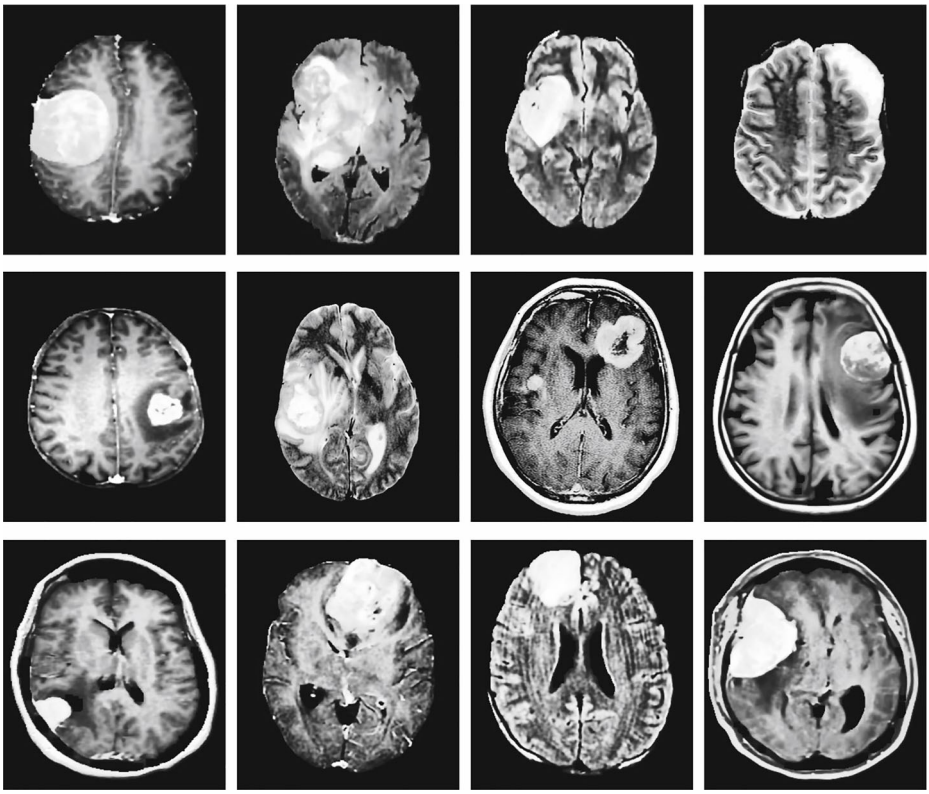

Fig. 3 Preprocessed images

Figure 3 demonstrates the preprocessing images. The FigShare dataset encompasses 3,064 brain MRI slices attained from 233 persons with three kinds of BT namely glioma, pituitary, and meningioma tumor. In this work, a set of 150 images are taken from each class. For experimental validation, the data set is divided into 30% and 70% of testing and training datasets.

4.3 Results analysis on Kaggle dataset

This section investigates the BT classification results of the MOAOA-FDL model on Kaggle dataset. Figure 4 reveals the confusion matrices of the MOAOA-FDL model on the test kaggle dataset. With 100 epochs, the MOAOA-FDL model has identified 228 and 143 images under abnormal and normal. Also, on 700 epochs, the MOAOA-FDL model has identified 232 and 144 images under abnormal and normal. At the same time, on 1200 epochs, the MOAOA-FDL model has categorized 232 and 142 images under abnormal and normal.

Table 1 depicts comprehensive BT classification outcomes of the MOAOA-FDL model on the test kaggle data set. The experimental value denotes that the MOAOA-FDL model has successfully categorized the images under every discrete epoch. For example, with 100 epochs, the MOAOA-FDL model has reached $sens_y$ of 97.28%, $spec_y$ of 97.85%, $accu_y$ of 97.63%, F_{score} of 96.95%, MCC of 95.02%, and kappa of 97.21%. Moreover, with 300 epochs, the MOAOA-FDL model has obtained $sens_y$ of 98.64%, $spec_y$ of 97.85%, $accu_y$ of

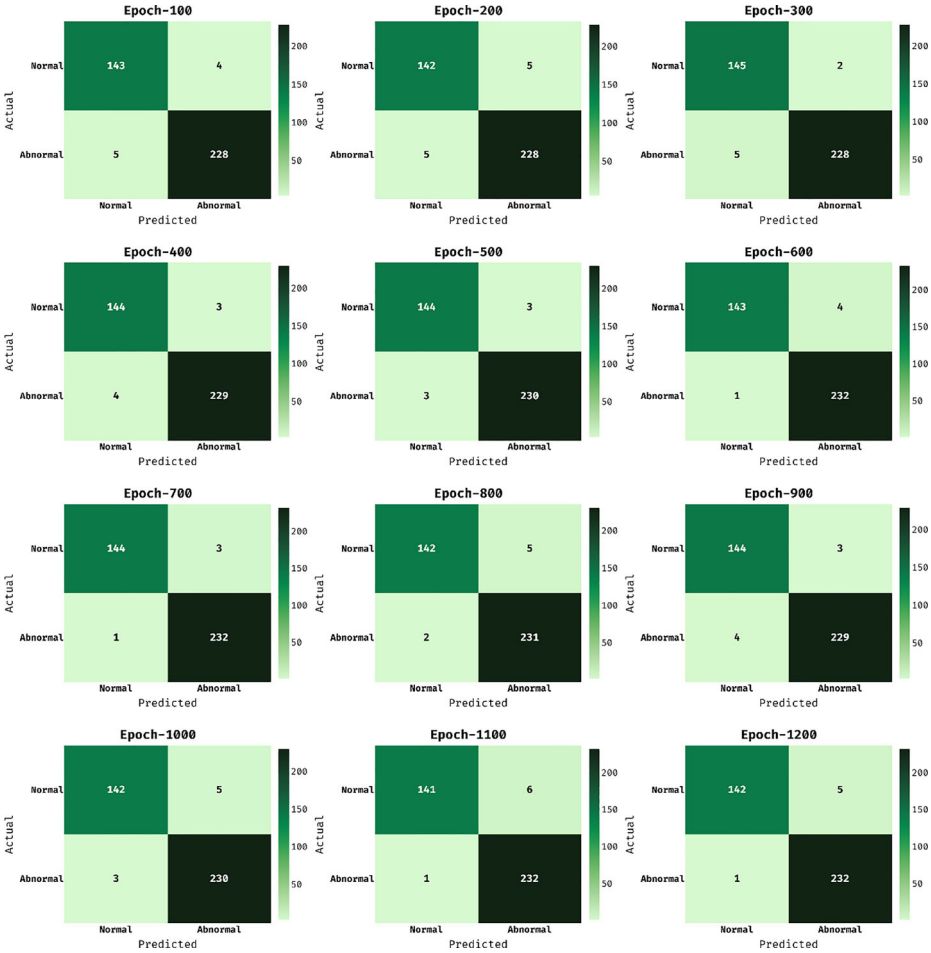

Fig. 4 Confusion matrix of MOAOA-FDL technique under kaggle datasets

Table 1 Performance analysis of MOAOA-FDL approach under kaggle data set

No. of Epochs	Sensitivity	Specificity	Accuracy	F-Score	MCC	Kappa
Epoch-100	97.28	97.85	97.63	96.95	95.02	97.21
Epoch-200	96.60	97.85	97.37	96.60	94.45	96.91
Epoch-300	98.64	97.85	98.16	97.64	96.14	97.83
Epoch-400	97.96	98.28	98.16	97.63	96.12	97.83
Epoch-500	97.96	98.71	98.42	97.96	96.67	98.14
Epoch-600	97.28	99.57	98.68	98.28	97.23	98.46
Epoch-700	97.96	99.57	98.95	98.63	97.78	98.77
Epoch-800	96.60	99.14	98.16	97.59	96.12	97.84
Epoch-900	97.96	98.28	98.16	97.63	96.12	97.83
Epoch-1000	96.60	98.71	97.89	97.26	95.56	97.53
Epoch-1100	95.92	99.57	98.16	97.58	96.13	97.85
Epoch-1200	96.60	99.57	98.42	97.93	96.68	98.15

97.37%, F_{score} of 96.60%, MCC of 96.14%, and kappa of 97.83%. Furthermore, with 1000 epochs, the MOAOA-FDL model has resulted to $sens_y$ of 96.60%, $spec_y$ of 98.71%, $accu_y$ of 97.89%, F_{score} of 97.26%, MCC of 95.56%, and kappa of 97.53%.

In Fig. 5 the accuracy investigation of the MOAOA-FDL technique under kaggle data set is reported. The figure indicated that the MOAOA-FDL approach has attained increased validation accuracy over the training accuracy. Also, it is indicated that the accuracy value gets saturated with the maximum epoch.

The loss outcome examination of the MOAOA-FDL approach under kaggle dataset is represented in Fig. 6. The figure depicted that the MOAOA-FDL method has signified the reduced validation loss over the training loss. Additionally, it is noted that the loss value gets saturated with the maximum epoch.

Followed by, a comprehensive study of the MOAOA-FDL method with existing appraoch takes place in Table 2. Figure 7 inspects $sens_y$ and $spec_y$ investigation of the MOAOA-FDL model with existing techniques on kaggle dataset. The results specified that the SVM-KP, SVM-RBF, CART, and k-NN models have demonstrated lower $accu_y$ values. Simultaneously, the DT and Linear SVM models have reached competitive results with $sens_y$ and $spec_y$ values. However, the MOAOA-FDL approach has obtained improved outcomes with the maximal $sens_y$ and $spec_y$ of 97.96%.

Figure 8 studies the acc_y analysis of the MOAOA-FDL approach with recent models on kaggle dataset. The figure reported that the k-NN, CART, RF, Linear SVM, and SVM-RBF models have exhibited reduced acc_y values of 80%, 84%, 88%, 88%, and 89.88% respectively. After that, the SVM-KP and DT models have resulted in near optimal outcomes with acc_y of 96.18% and 94%. However, the MOAOA-FDL model has outperformed the other technique with an increased *kappa* of 98.95%

A detailed computation time (CT) examination of the MOAOA-FDL model with existing models on the Kaggle dataset is shown in Fig. 9. The figure inferred that the SVM-RBF and DT models have reported higher CT of 12.05s and 11.43s respectively. Along with that, the

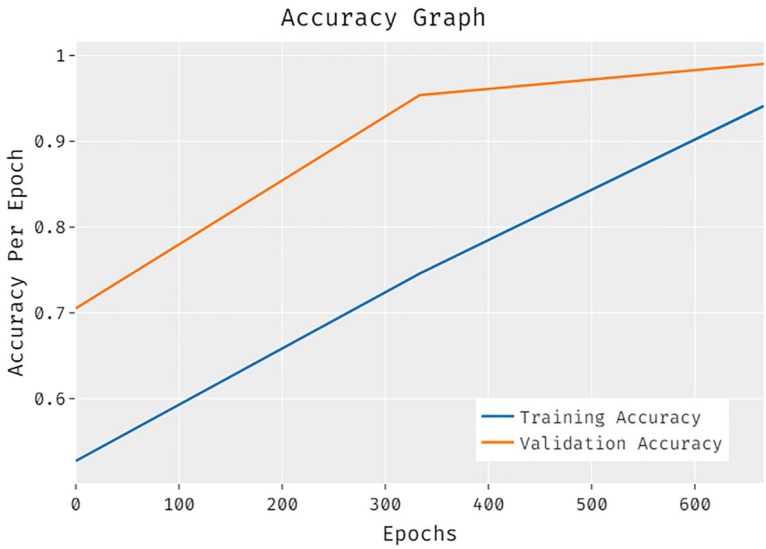

Fig. 5 Accuracy analysis of MOAOA-FDL approach under Kaggle data set

Fig. 6 Loss analysis of MOAOA-FDL method under Kaggle dataset

CART, SVM-KP, RF, and k-NN models have shown slightly reduced CT of 9.44s, 7.56s, 6.67s, and 8.06s respectively. Followed by, the DenseNet201 and MobileNet v2 models have accomplished reasonable CT of 6.76 and 6.92 s respectively. Though the linear SVM model has resulted to reasonable CT of 5.22s, the MOAOA-FDL model has accomplished superior performance with lower CT of 4.32s.

4.4 Results analysis on Figshare dataset

In this section, the BT detection and classification results on Figshare dataset are discussed. Figure 10 demonstrates the confusion matrices of the MOAOA-FDL model on the test Figshare dataset. On 100 epochs, the MOAOA-FDL method has recognized 224 images under Glioma class, 223 images under Pituitary class, and 233 images under Meningioma class. Also, on 700 epochs, the MOAOA-FDL model has categorized 223 images under Glioma class, 223 images under Pituitary class, and 220 images under Meningioma class. At the same time, on 1200 epochs, the MOAOA-FDL method has identified 224 images under Glioma class, 225 images under Pituitary class, and 224 images under Meningioma class.

Table 2 Comparative Analysis of Existing with Proposed Methodology interms of Sensitivity, Specificity, Accuracy on Kaggle dataset

Methods	Sensitivity	Specificity	Accuracy	Computational Time (Sec)
MOAOA-FDL	97.96	99.57	98.95	4.32
SVM-KP Model	94.73	97.59	96.18	7.56
SVM-RBF Model	95.62	83.71	89.88	12.05
Decision Tree Algorithm	97.88	91.71	94.95	11.43
CART Algorithm	88.00	80.00	84.00	9.44
Random Forest	96.00	80.00	88.00	6.97
k-NN Algorithm	80.00	80.00	80.00	8.06
Linear SVM Model	96.00	80.00	88.00	5.22

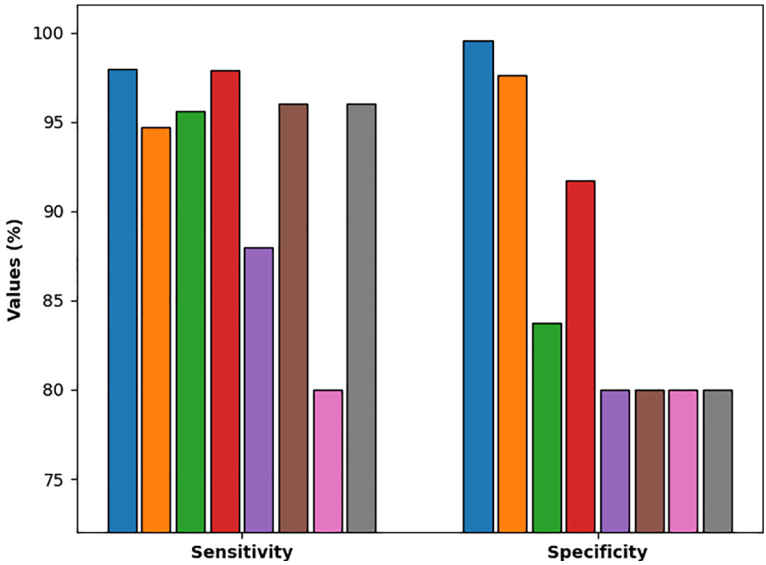

Fig. 7 $Sens_y$ and $Spec_y$ analysis of MOAOA-FDL approach under kaggle data set

Table 3 provides comprehensive BT classification outcomes of the MOAOA-FDL model under discrete classes and epochs. The result shows that the MOAOA-FDL technique has accomplished effectual classification outcomes under all classes and epochs.

Table 4 portrays the overall BT classification results of the MOAOA-FDL method on the test Figshare data set. The result shows that the MOAOA-FDL technique has effectually classified the images under all distinct epochs. For instance, with 100 epochs, the MOAOA-FDL model has accomplished $sens_y$ of 99.26%, $spec_y$ of 99.63%, $accu_y$ of 99.51%, F_{score} of 99.26%, MCC of 98.89%, and kappa of 98.89%. At the same time, with 300 epochs, the

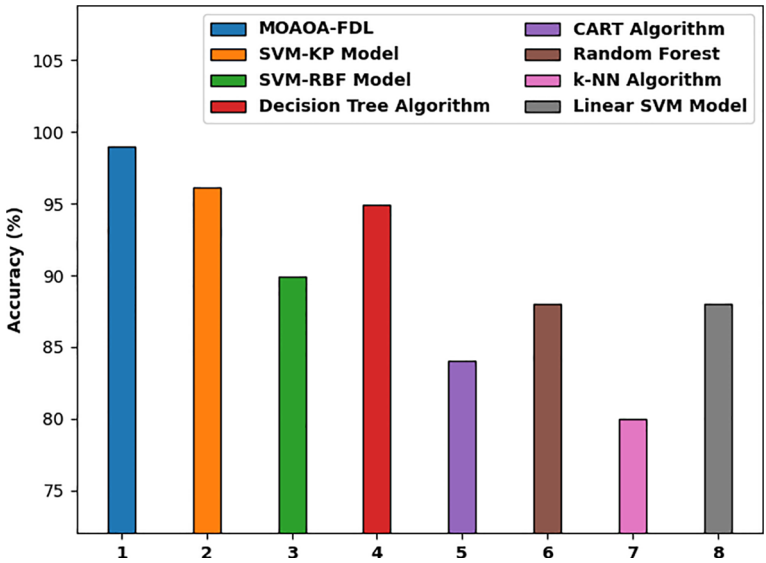

Fig. 8 Acc_y analysis of MOAOA-FDL approach under kaggle data set

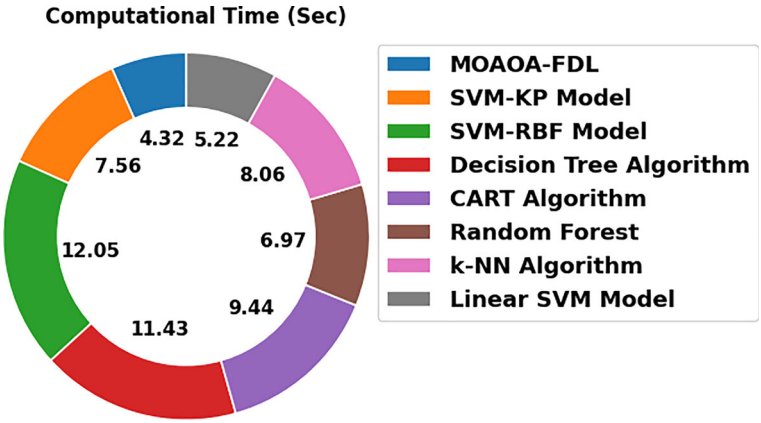

Fig. 9 CT analysis of MOAOA-FDL approach under kaggle data set

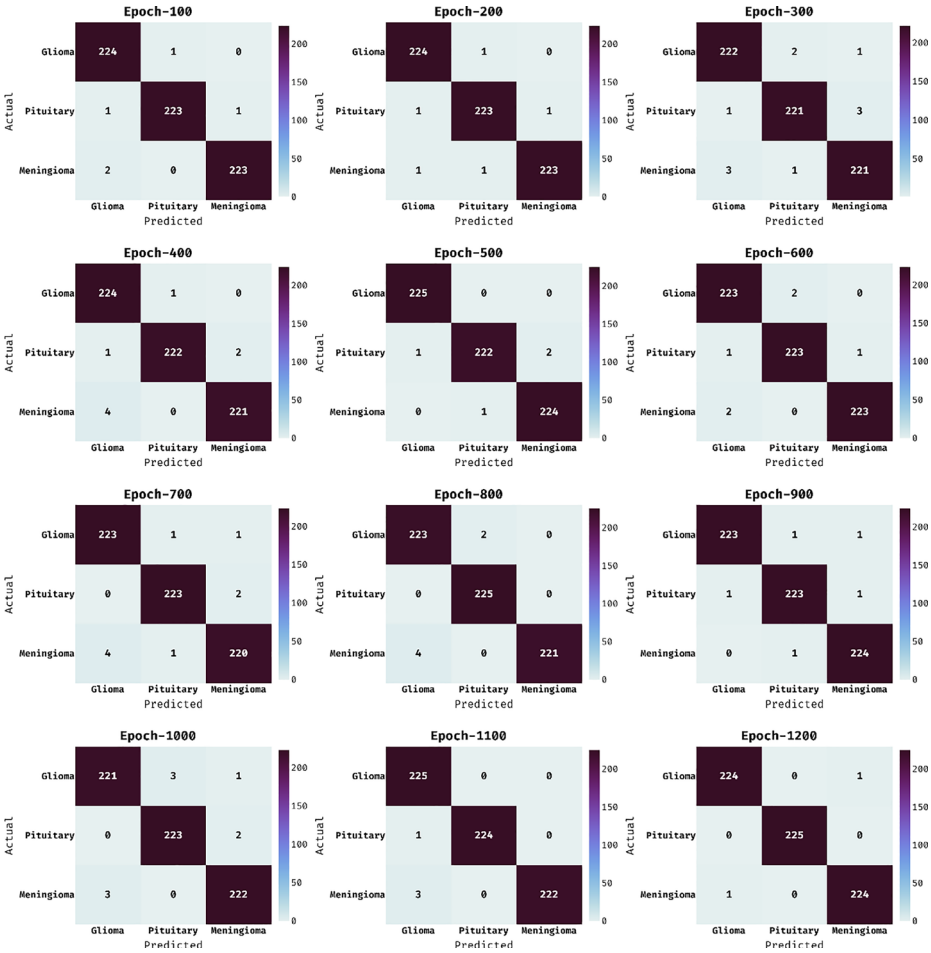

Fig. 10 Confusion matrix of MOAOA-FDL technique under Figshare dataset

Table 3 Result analysis of MOAOA-FDL model under distinct classes and epochs

Methods	Sensitivity	Specificity	Accuracy	F-Score	MCC	Kappa
Epoch-100						
Glioma	99.56	99.33	99.41	99.12	98.67	-
Pituitary	99.11	99.78	99.56	99.33	99.00	-
Meningioma	99.11	99.78	99.56	99.33	99.00	-
Average	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-200						
Glioma	99.56	99.56	99.56	99.33	99.00	-
Pituitary	99.11	99.56	99.41	99.11	98.67	-
Meningioma	99.11	99.78	99.56	99.33	99.00	-
Average	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-300						
Glioma	98.67	99.11	98.96	98.45	97.67	-
Pituitary	98.22	99.33	98.96	98.44	97.66	-
Meningioma	98.22	99.11	98.81	98.22	97.33	-
Average	98.37	99.19	98.91	98.37	97.56	97.56
Epoch-400						
Glioma	99.56	98.89	99.11	98.68	98.02	-
Pituitary	98.67	99.78	99.41	99.11	98.67	-
Meningioma	98.22	99.56	99.11	98.66	98.00	-
Average	98.81	99.41	99.21	98.82	98.23	98.22
Epoch-500						
Glioma	100.00	99.78	99.85	99.78	99.67	-
Pituitary	98.67	99.78	99.41	99.11	98.67	-
Meningioma	99.56	99.56	99.56	99.33	99.00	-
Average	99.41	99.70	99.60	99.41	99.11	99.11
Epoch-600						
Glioma	99.11	99.33	99.26	98.89	98.34	-
Pituitary	99.11	99.56	99.41	99.11	98.67	-
Meningioma	99.11	99.78	99.56	99.33	99.00	-
Average	99.11	99.56	99.41	99.11	98.67	98.67
Epoch-700						
Glioma	99.11	99.11	99.11	98.67	98.01	-
Pituitary	99.11	99.56	99.41	99.11	98.67	-
Meningioma	97.78	99.33	98.81	98.21	97.33	-
Average	98.67	99.33	99.11	98.67	98.00	98.00
Epoch-800						
Glioma	99.11	99.11	99.11	98.67	98.01	-
Pituitary	100.00	99.56	99.70	99.56	99.34	-
Meningioma	98.22	100.00	99.41	99.10	98.67	-
Average	99.11	99.56	99.41	99.11	98.67	98.67
Epoch-900						
Glioma	99.11	99.78	99.56	99.33	99.00	-
Pituitary	99.11	99.56	99.41	99.11	98.67	-
Meningioma	99.56	99.56	99.56	99.33	99.00	-
Average	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-1000						
Glioma	98.22	99.33	98.96	98.44	97.66	-
Pituitary	99.11	99.33	99.26	98.89	98.34	-
Meningioma	98.67	99.33	99.11	98.67	98.00	-
Average	98.67	99.33	99.11	98.67	98.00	98.00
Epoch-1100						
Glioma	100.00	99.11	99.41	99.12	98.68	-
Pituitary	99.56	100.00	99.85	99.78	99.67	-
Meningioma	98.67	100.00	99.56	99.33	99.00	-
Average	99.41	99.70	99.60	99.41	99.12	99.11
Epoch-1200						
Glioma	99.56	99.78	99.70	99.56	99.33	-

Table 3 (continued)

Methods	Sensitivity	Specificity	Accuracy	F-Score	MCC	Kappa
Pituitary	100.00	100.00	100.00	100.00	100.00	-
Meningioma	99.56	99.78	99.70	99.56	99.33	-
Average	99.70	99.85	99.80	99.70	99.56	99.56

MOAOA-FDL model has achieved $sens_y$ of 98.37%, $spec_y$ of 99.19%, $accu_y$ of 98.91%, F_{score} of 98.37%, MCC of 97.56%, and kappa of 97.56%. In line with, with 1000 epochs, the MOAOA-FDL model has attained $sens_y$ of 98.67%, $spec_y$ of 99.33%, $accu_y$ of 99.11%, F_{score} of 98.67%, MCC of 98%, and kappa of 98%.

In Fig. 11, the accuracy outcome examination of the MOAOA-FDL approach under Figshare dataset is portrayed. The results established that the MOAOA-FDL method has depicted better validation accuracy over the training accuracy. It is noted that the accuracy value is increased with rising epochs.

The loss outcome of the MOAOA-FDL methodology under Figshare dataset is depicted in Fig. 12. The figure revealed that the MOAOA-FDL approach has signified the reduced validation loss over the training loss. Additionally, it is noted that the loss value gets saturated with increased epochs.

Finally, a brief comparison of the MOAOA-FDL technique with existing models takes place in Table 5 [6]. Figure 13 examines the $accu_y$ analysis of the MOAOA-FDL method with current techniques on Figshare dataset. The outcome indicates that the DenseNet201, ResNet50, Inception v3, and MobileNetv2 models have demonstrated lesser $accu_y$ values of 93.10%, 92.90%, 92.80%, and 91.80% correspondingly. Simultaneously, the NASNet technique has resulted in near optimal outcomes with the $accu_y$ of 99.60%. However, the MOAOA-FDL model has resulted in enhanced performance with the maximum $accu_y$ of 99.80%.

Figure 14 examines the *kappa* analysis of the MOAOA-FDL approach with existing models on Figshare dataset. The figure reported that the DenseNet201, ResNet50, Inception v3, and MobileNetv2 models have demonstrated lower *kappa* values of 89%, 89%, 88%, and 87% respectively. In line with this, the NASNet model has accomplished a competitive outcome with the *kappa* of 99%. However, the MOAOA-FDL technique has depicted better outcome with an increased *kappa* of 99.56%

Table 4 Overall BT classification analysis of MOAOA-FDL model under Figshare dataset

No. of Epochs	Sensitivity	Specificity	Accuracy	F-Score	MCC	Kappa
Epoch-100	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-200	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-300	98.37	99.19	98.91	98.37	97.56	97.56
Epoch-400	98.81	99.41	99.21	98.82	98.23	98.22
Epoch-500	99.41	99.70	99.60	99.41	99.11	99.11
Epoch-600	99.11	99.56	99.41	99.11	98.67	98.67
Epoch-700	98.67	99.33	99.11	98.67	98.00	98.00
Epoch-800	99.11	99.56	99.41	99.11	98.67	98.67
Epoch-900	99.26	99.63	99.51	99.26	98.89	98.89
Epoch-1000	98.67	99.33	99.11	98.67	98.00	98.00
Epoch-1100	99.41	99.70	99.60	99.41	99.12	99.11
Epoch-1200	99.70	99.85	99.80	99.70	99.56	99.56

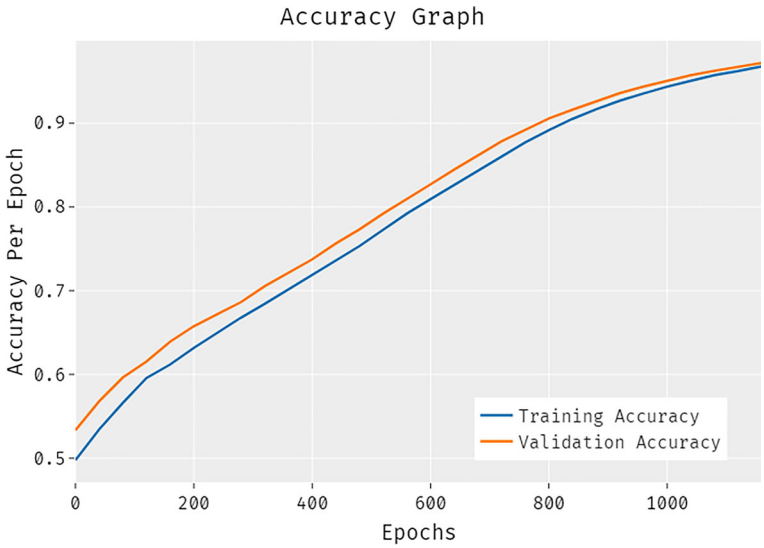

Fig. 11 Accuracy analysis of MOAOA-FDL technique under Figshare dataset

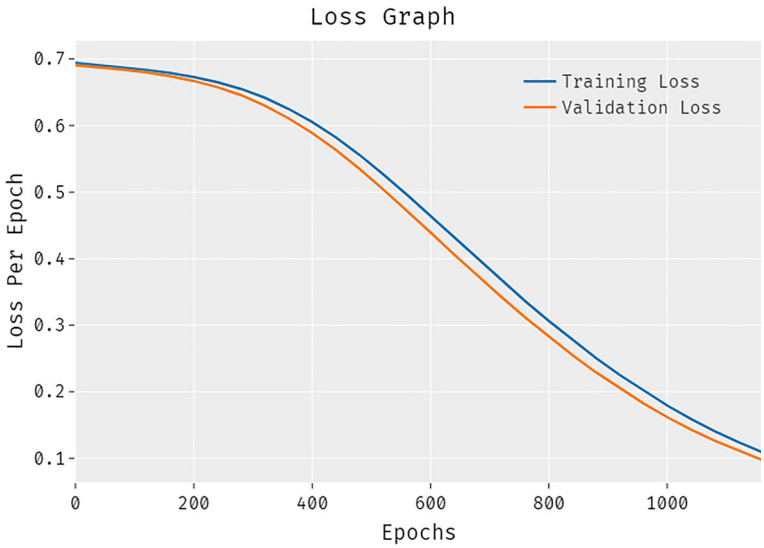

Fig. 12 Loss analysis of MOAOA-FDL model under Figshare dataset

Table 5 Comparative analysis of MOAOA-FDL approach with recent techniques under Figshare dataset

Methods	Accuracy	Kappa	Computational Time (Sec)
MOAOA-FDL	99.80	99.56	5.20
NASNet	99.60	99.00	8.06
DenseNet201	93.10	89.00	6.76
ResNet50	92.90	89.00	13.32
Inception V3	92.80	88.00	10.98
MobileNet V2	91.80	87.00	6.92

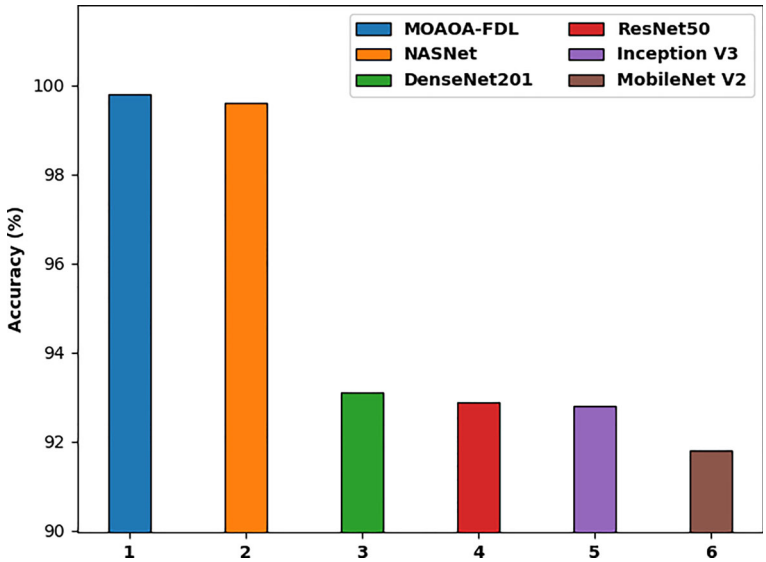

Fig. 13 Acc_y analysis of MOAOA-FDL technique with recent approaches under Figshare dataset

Figure 15 provides a comparative CT analysis of the MOAOA-FDL model with existing models on the Figshare dataset. The figure implied that the ResNet50 and Inception v3 models have reported higher CT of 13.32s and 10.98s respectively. Along with that, the NASNet model has shown slightly reduced CT of 8.06s. Followed by, the DenseNet201 and MobileNet v2 models have accomplished reasonable CT of 6.76 and 6.92 s respectively. However, the MOAOA-FDL model has resulted to effectual outcome with minimal CT of 5.20s.

By observing into the abovementioned results and discussion, it is ensured that the MOAOA-FDL model has obtained maximal BT detection and classification performance.

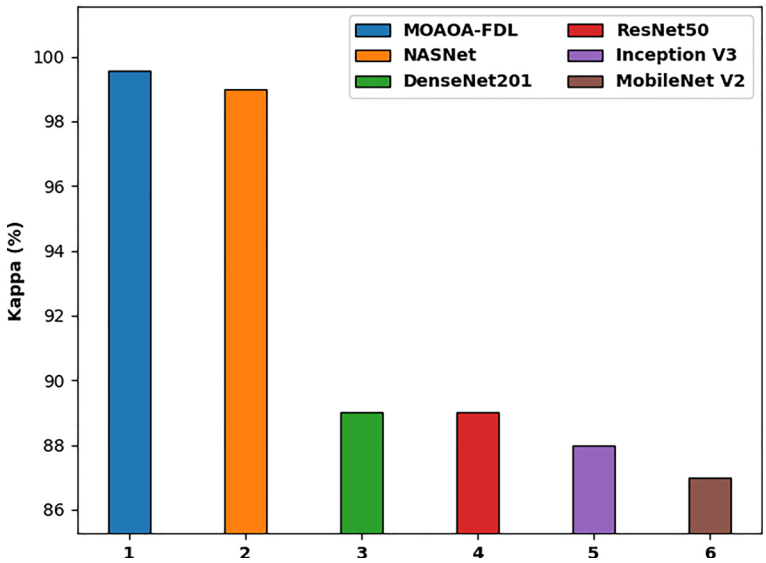

Fig. 14 *Kappa* analysis of MOAOA-FDL technique with recent approaches under Figshare dataset

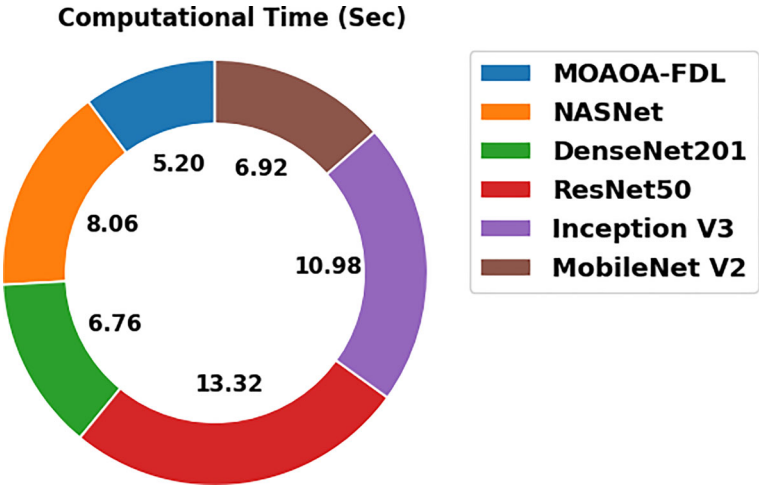

Fig. 15 CT analysis of MOAOA-FDL technique with recent approaches under Figshare dataset

The MOAOA-FDL model has shown superior performance due to the inclusion of AOA algorithm for the parameter selection of the LSTM model and threshold value selection of the Shannon entropy process.

5 Conclusion

In this study, a new MOAOA-FDL approach has been developed for the detection and classification of BT with MRI images. The suggested MOAOA-FDL models incorporate preprocessing, Shannon's entropy based segmentation, fusion enabled feature extraction, LSTM based classification, and AOA based hyperparameter optimization. In order to illustrate the better performance of the MOAOA-FDL method, a series of simulations were performed using benchmark dataset. The experimental result shows that the MOAOA-FDL technique has outperformed the other recent techniques with maximum accuracy of 98.95% and 99.80% on the applied Kaggle and Figshare datasets respectively. Therefore, the MOAOA-FDL technique can accomplish maximum BT classification performance. In the future, the proposed MOAOA-FDL technique can be extended to the design of DL based instance segmentation approach. Besides, the proposed model can be tested using large scale real time datasets in future.

Declarations

Human participants and/or animals Not applicable.

Conflict of interest The authors have expressed no conflict of interest.

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